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EN.625.633.81
Homework N

1.

We are given the stochastic differential equation for the Ornstein-Uhlenbrek process:

$$dX(t) = \theta(\mu - X(t))dt + \sigma dW(t)$$

with the analytical solution as

$$X(t) = X(0)e^{-\theta t} + \mu(1 - e^{-\theta t}) + \int_0^t \sigma e^{\theta(s-t)} dW(s)$$

2.

We aim to illustrate through R the difference between the Ito ingeral and the Stratonovich integral; i.e. find some value of $h(t)$ and simulated process $W(t)$ so that calculating the integral $\int_0^T h(t)dW(t)$ is considerably different for the two calculations.

`asdf`

3.

The approximation given in (5.4) of the paper is

$$\log e_{\Delta t}^{\text{strong}} \approx \log C + \frac{1}{2} \log \Delta t$$

so we try to illustrate this by plotting the resulting graph of average error vs. Δt on a log scale, where we should see an approximately straight line with a slope of $\frac{1}{2}$. We modify the function slightly to return $averageFinalDifference$, and then run accordingly and graph the results:

```
> T_values = c(0.25, 0.5, 1, 2, 5, 10)
> N_values = 2^(2:10)
> # Same as paper
> lambda = 2
> mu = 1
> nmbSimulations = 1000
```

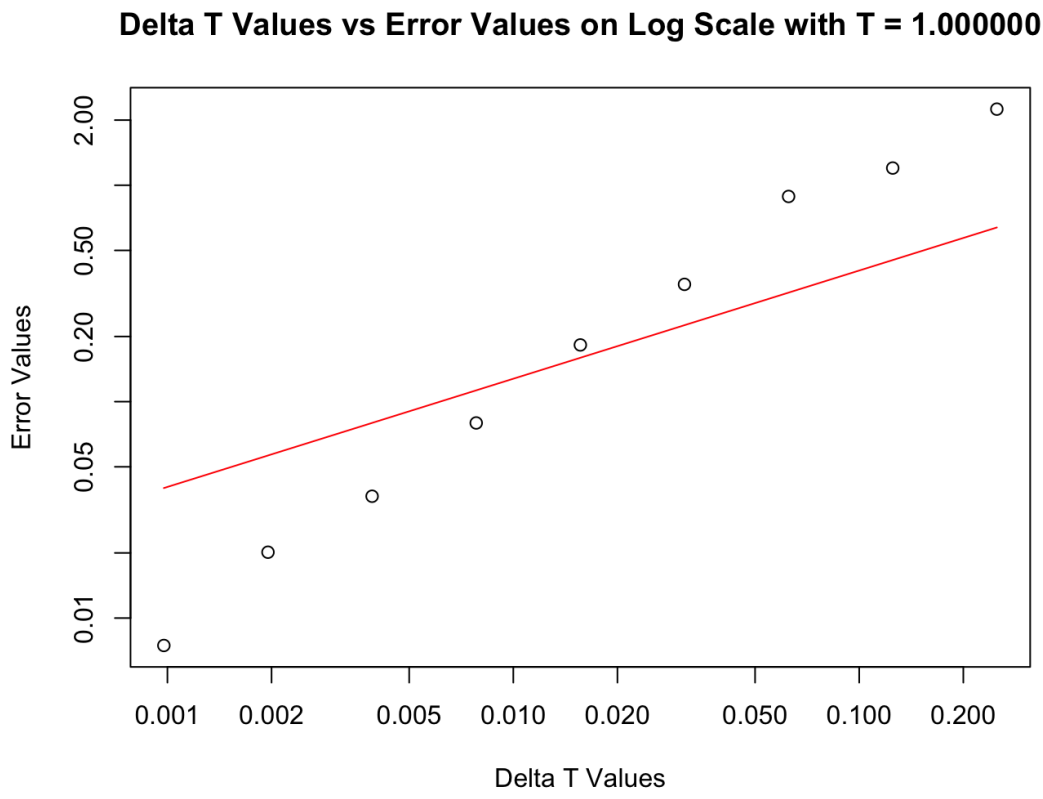
```
>
> for (T in T_values) {
+   deltat_values = as.numeric(list())
+   error_values = as.numeric(list())
+
+   for (N in N_values) {
+     sprintf("running for T = %f, N = %d...", T, N)
+
+     deltat_values = append(deltat_values, T / N)
+     error_values = append(error_values, abs(blackScholesConvergence(T,
N, lambda, mu, nmbSimulations)))
+   }
+
+   plot(
+     deltat_values,
+     error_values,
+     log = "xy",
+     xlab = "Delta T Values",
+     ylab = "Error Values",
+     main = sprintf("Delta T Values vs Error Values on Log Scale with
T = %f", T)
+   )
+
+   # Reference line with exact slope as 1/2 fitted to similar offset
+   ref_C = exp(mean(log(error_values)) - 0.5 * mean(log(deltat_values)))
+   y = ref_C * sqrt(deltat_values)
+   lines(
+     deltat_values,
+     y,
+     type="l",
```

```
+     col = "red",
+   )
+ }
[1] "average deviation =  0.0492247166559775 , and the delta t =  0.0625"
[1] "average deviation =  0.024091028137924 , and the delta t =  0.03125"
[1] "average deviation =  0.0137789607057976 , and the delta t =  0.015625"
[1] "average deviation =  0.00560154546568383 , and the delta t =  0.0078125
[1] "average deviation =  0.00461061928605198 , and the delta t =  0.0039062
[1] "average deviation =  0.00194927934579275 , and the delta t =  0.0019531
[1] "average deviation =  0.000323983109936274 , and the delta t =  0.000976
[1] "average deviation =  0.00126783021087313 , and the delta t =  0.0004882
[1] "average deviation =  0.000250399393489314 , and the delta t =  0.00024
[1] "average deviation =  0.31711713984836 , and the delta t =  0.125"
[1] "average deviation =  0.132171536010918 , and the delta t =  0.0625"
[1] "average deviation =  0.0743188255687851 , and the delta t =  0.03125"
[1] "average deviation =  0.051269002108907 , and the delta t =  0.015625"
[1] "average deviation =  0.0197699304179946 , and the delta t =  0.0078125"
[1] "average deviation =  0.0150187575230333 , and the delta t =  0.00390625
[1] "average deviation =  0.00449707590957639 , and the delta t =  0.0019531
[1] "average deviation =  0.00146653530308295 , and the delta t =  0.000976
[1] "average deviation =  0.000595854856125777 , and the delta t =  0.00048
[1] "average deviation =  2.2471645677081 , and the delta t =  0.25"
[1] "average deviation =  1.20059710406355 , and the delta t =  0.125"
[1] "average deviation =  0.887552190793862 , and the delta t =  0.0625"
[1] "average deviation =  0.348304934738151 , and the delta t =  0.03125"
[1] "average deviation =  0.182839774132024 , and the delta t =  0.015625"
[1] "average deviation =  0.0797147803555463 , and the delta t =  0.0078125"
[1] "average deviation =  0.0365496282298914 , and the delta t =  0.00390625
[1] "average deviation =  0.0201513669795156 , and the delta t =  0.00195312
[1] "average deviation =  0.00747017151247117 , and the delta t =  0.000976
```

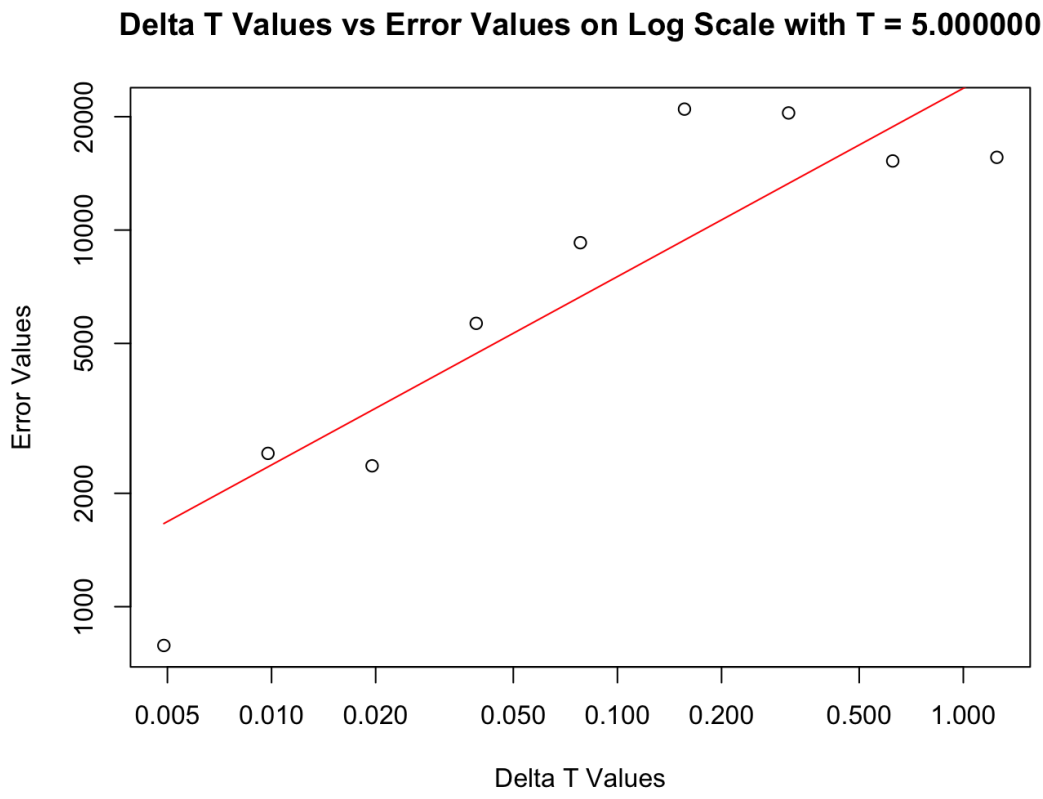
```
[1] "average deviation =  38.3860350148055 , and the delta t =  0.5"
[1] "average deviation =  31.8416907859503 , and the delta t =  0.25"
[1] "average deviation =  21.4146284433919 , and the delta t =  0.125"
[1] "average deviation =  14.7718342858946 , and the delta t =  0.0625"
[1] "average deviation =  5.56539348123626 , and the delta t =  0.03125"
[1] "average deviation =  3.41716607124377 , and the delta t =  0.015625"
[1] "average deviation =  1.41197509467434 , and the delta t =  0.0078125"
[1] "average deviation =  1.2757240639073 , and the delta t =  0.00390625"
[1] "average deviation =  0.24428420763604 , and the delta t =  0.001953125"
[1] "average deviation =  15597.69661595 , and the delta t =  1.25"
[1] "average deviation =  15255.3191487282 , and the delta t =  0.625"
[1] "average deviation =  20462.1594433599 , and the delta t =  0.3125"
[1] "average deviation =  20943.8653605296 , and the delta t =  0.15625"
[1] "average deviation =  9256.47853698749 , and the delta t =  0.078125"
[1] "average deviation =  5654.01247705172 , and the delta t =  0.0390625"
[1] "average deviation =  2366.18215557262 , and the delta t =  0.01953125"
[1] "average deviation =  2552.93685449814 , and the delta t =  0.009765625"
[1] "average deviation =  788.906730669276 , and the delta t =  0.0048828125
[1] "average deviation =  287769024.071099 , and the delta t =  2.5"
[1] "average deviation =  513211342.45659 , and the delta t =  1.25"
[1] "average deviation =  362676431.210016 , and the delta t =  0.625"
[1] "average deviation =  364272514.331549 , and the delta t =  0.3125"
[1] "average deviation =  265851877.484744 , and the delta t =  0.15625"
[1] "average deviation =  229631572.665997 , and the delta t =  0.078125"
[1] "average deviation =  161346784.854953 , and the delta t =  0.0390625"
[1] "average deviation =  254194191.496466 , and the delta t =  0.01953125"
[1] "average deviation =  36543557.0686745 , and the delta t =  0.009765625"
```

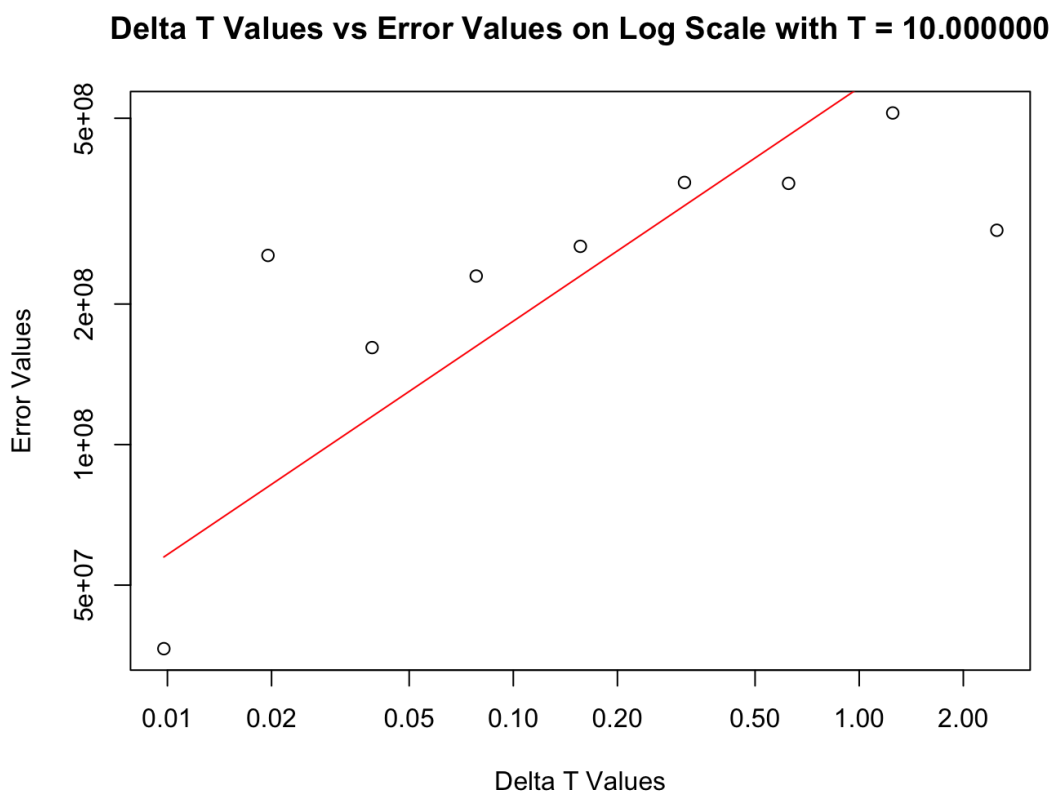
Delta T Values vs Error Values on Log Scale with T = 0.250000
Error Values
5e-04
2e-03
5e-03
2e-02
5e-02
2e-04
5e-04
1e-03
2e-03
5e-03
1e-02
2e-02
5e-02
Delta T Values

Delta T Values vs Error Values on Log Scale with T = 0.500000
Error Values
5e-04
2e-03
1e-02
5e-02
2e-01
5e-04
1e-03
2e-03
5e-03
1e-02
2e-02
5e-02
1e-01
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 1.000000
Error Values
2.00
0.50
0.20
0.05
0.01
0.001
0.002
0.005
0.010
0.020
0.050
0.100
0.200
Delta T Values

Delta T Values vs Error Values on Log Scale with T = 2.000000
Error Values
20.0
10.0
5.0
2.0
1.0
0.5
0.2
0.002
0.005
0.010
0.020
0.050
0.100
0.200
0.500
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 5.000000
Error Values
1000
2000
5000
10000
20000
0.005
0.010
0.020
0.050
0.100
0.200
0.500
1.000
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 10.000000
Error Values
5e+07
1e+08
2e+08
5e+08
0.01
0.02
0.05
0.10
0.20
0.50
1.00
2.00
Delta T Values

We plotted the adjusted lines with slope of $\frac{1}{2}$ and we do see that generally, the results do follow the approximation given in the paper. At $T = 10$, we begin to falter a bit more, but it is pretty close for all the other values for T.